Lecture 17 (*March 31*)

Reading: Section 8.2
Viewing: Lecture 31

In our last lecture, we said that a linear transformation $\phi : \mathbb{R}^n \to \mathbb{R}^m$ corresponds to an $m \times n$ matrix A. However, this is not completely precise: while a linear transformation ϕ is an intrinsic geometric notion (namely a rule that sends points to points and lines to lines) the matrix A is simply a representation of it. For example, think of elements of $\mathbb{R}^2$ as points in the plane, and consider the linear transformation:

$$\phi : \mathbb{R}^2 \to \mathbb{R}^2$$

given by the following rule for all points p in the plane:

$$\phi(p) = (p \text{ rotated counterclockwise by } 30^\circ)$$

The matrix $A = \frac{1}{2}\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$ is a representation of this rotation in the usual (x, y) system of coordinates, i.e. in the usual basis $\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ of $\mathbb{R}^2$. Indeed, all that this is saying is that:

$$\phi(x \cdot \boldsymbol{e}_1 + y \cdot \boldsymbol{e}_2) = x \cdot \boldsymbol{a}_1 + y \cdot \boldsymbol{a}_2 = \left(\frac{x\sqrt{3}}{2} - \frac{y}{2}\right) \cdot \boldsymbol{e}_1 + \left(\frac{x}{2} + \frac{y\sqrt{3}}{2}\right) \cdot \boldsymbol{e}_2 \tag{138}$$

where $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ are the columns of A. But now suppose you wanted to express this same geometric transformation in a different reference frame, i.e. changing the basis from $\boldsymbol{e}_1, \boldsymbol{e}_2$ to:

$$\boldsymbol{v}_1 = 2\boldsymbol{e}_1, \quad \boldsymbol{v}_2 = \boldsymbol{e}_1 - \boldsymbol{e}_2.$$

When you change basis from $\boldsymbol{e}$'s to $\boldsymbol{v}$'s, you can always change back, i.e. express the $\boldsymbol{e}$'s in terms of $\boldsymbol{v}$'s. In the case above, we have (solving for $\boldsymbol{e}$'s in terms of $\boldsymbol{v}$'s):

$$\boldsymbol{e}_1 = \frac{\boldsymbol{v}_1}{2}, \quad \boldsymbol{e}_2 = \frac{\boldsymbol{v}_1}{2} - \boldsymbol{v}_2$$

We may plug these formulas into (138) and we get:

$$\phi(x \cdot \boldsymbol{v}_1 + y \cdot \boldsymbol{v}_2) = \phi(x \cdot 2\boldsymbol{e}_1 + y \cdot (\boldsymbol{e}_1 - \boldsymbol{e}_2)) = \phi((2x + y) \cdot \boldsymbol{e}_1 - y \cdot \boldsymbol{e}_2) \overset{(138)}{=} \tag{139}$$

$$\left[\frac{(2x+y)\sqrt{3}}{2} + \frac{y}{2}\right]\boldsymbol{e}_1 + \left[\frac{2x+y}{2} - \frac{y\sqrt{3}}{2}\right]\boldsymbol{e}_2 = \left[\frac{(2x+y)\sqrt{3}}{2} + \frac{y}{2}\right]\frac{\boldsymbol{v}_1}{2} + \left[\frac{2x+y}{2} - \frac{y\sqrt{3}}{2}\right]\left(\frac{\boldsymbol{v}_1}{2} - \boldsymbol{v}_2\right)$$

$$= \left[x \cdot \frac{\sqrt{3}+1}{2} + y \cdot \frac{1}{2}\right] \cdot \boldsymbol{v}_1 + \left[x \cdot (-1) + y \cdot \frac{\sqrt{3}-1}{2}\right] \cdot \boldsymbol{v}_2$$

So in terms of the new basis $\boldsymbol{v}_1, \boldsymbol{v}_2$, the linear transformation ϕ is represented by the matrix:

$$B = \begin{bmatrix} \frac{\sqrt{3}+1}{2} & \frac{1}{2} \\ -1 & \frac{\sqrt{3}-1}{2} \end{bmatrix}$$

The connection between this matrix and the original matrix A is precisely:

$$\boxed{B = V^{-1}AV} \tag{140}$$

where $V = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$ is the matrix whose columns are the vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ written in the $\boldsymbol{e}_1, \boldsymbol{e}_2$ basis, i.e.:

$$V\boldsymbol{e}_1 = \boldsymbol{v}_1 \quad \text{and} \quad V\boldsymbol{e}_2 = \boldsymbol{v}_2 \tag{141}$$

In other words, V is the matrix whose columns are the vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. We call V the:

$\boxed{\textbf{change of basis matrix}}$ **from the basis $\boldsymbol{v}_1, \boldsymbol{v}_2$ to the basis $\boldsymbol{e}_1, \boldsymbol{e}_2$**

Remark. *You might be a bit surprised that we call* (141) *the "change of basis matrix <u>from</u> the basis $\boldsymbol{v}_1, \boldsymbol{v}_2$ <u>to</u> the basis $\boldsymbol{e}_1, \boldsymbol{e}_2$" and not vice-versa. This is a terminology issue, but let me try to justify it. Take a random vector, say $\boldsymbol{v} = 2\boldsymbol{v}_1 + 3\boldsymbol{v}_2$. In the basis $\boldsymbol{v}_1, \boldsymbol{v}_2$:*

$$\boldsymbol{v} \quad \textit{is represented by} \quad \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

But in the standard basis $\boldsymbol{e}_1, \boldsymbol{e}_2$, since $\boldsymbol{v}_1 = V\boldsymbol{e}_1$ and $\boldsymbol{v}_2 = V\boldsymbol{e}_2$:

$$\boldsymbol{v} \quad \textit{is represented by} \quad V \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

To summarize: if an arbitrary vector $\boldsymbol{v} \in \mathbb{R}^2$ is represented by a column in the basis $\boldsymbol{v}_1, \boldsymbol{v}_2$, then it is represented by V times that column in the basis $\boldsymbol{e}_1, \boldsymbol{e}_2$. Hence the "...<u>from</u>...<u>to</u>..." terminology.

So we have seen that the matrices A and B connected by (140) describe the same linear transformation, albeit in different bases. Let us explain the general principle for any linear transformation:

$$\phi : \mathbb{R}^n \to \mathbb{R}^n$$

which is represented by the $n \times n$ matrix $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$ in the standard basis:

$$\phi(x_1\boldsymbol{e}_1 + \cdots + x_n\boldsymbol{e}_n) = (a_{11}x_1 + \cdots + a_{1n}x_n)\boldsymbol{e}_1 + \cdots + (a_{n1}x_1 + \cdots + a_{nn}x_n)\boldsymbol{e}_n \tag{142}$$

If you wish to express the linear transformation ϕ in a different basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$, then you need to consider the **similar** matrix:

$$B = V^{-1}AV = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \dots & b_{nn} \end{bmatrix}$$

where $V = [\ \boldsymbol{v}_1 \mid \dots \mid \boldsymbol{v}_n\ ]$ is the change of basis matrix from the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$. In terms of the $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ basis, formula (142) may be rewritten as:

$$\phi(x_1\boldsymbol{v}_1 + \cdots + x_n\boldsymbol{v}_n) = (b_{11}x_1 + \cdots + b_{1n}x_n)\boldsymbol{v}_1 + \cdots + (b_{n1}x_1 + \cdots + b_{nn}x_n)\boldsymbol{v}_n \tag{143}$$

Example 5. *The projection onto the vector* $\begin{bmatrix}1\\1\end{bmatrix}$ *in* $\mathbb{R}^2$ *is represented by the matrix:*

$$A = \begin{bmatrix}1\\1\end{bmatrix}\left(\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}\right)^{-1}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}$$

in the usual $\boldsymbol{e}_1, \boldsymbol{e}_2$ *basis of the plane. But suppose you wanted to represent the same linear transformation in the basis* $\boldsymbol{v}_1 = 2\boldsymbol{e}_1 + 2\boldsymbol{e}_2$, $\boldsymbol{v}_2 = 3\boldsymbol{e}_1 - 3\boldsymbol{e}_2$ *instead. To do so, consider the change of basis matrix:*

$$V = \begin{bmatrix}2 & 3\\ 2 & -3\end{bmatrix}$$

Then formula (140) *says that the given projection linear transformation is represented by the matrix:*

$$B = V^{-1}AV = \begin{bmatrix}\frac{1}{4} & \frac{1}{4}\\ \frac{1}{6} & -\frac{1}{6}\end{bmatrix}\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}\begin{bmatrix}2 & 3\\ 2 & -3\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$$

in the new $\boldsymbol{v}_1, \boldsymbol{v}_2$ *basis. It's not a coincidence that the answer is such a simple matrix. Indeed, all that the formula for* B *is saying is that:*

$$\operatorname{proj}_{\begin{bmatrix}1\\1\end{bmatrix}}(x \cdot \boldsymbol{v}_1 + y \cdot \boldsymbol{v}_2) = x \cdot \boldsymbol{v}_1$$

Geometrically, this says that projection sends $\boldsymbol{v}_1$ *to itself (i.e. leaves it unchanged, which makes sense given that* $\boldsymbol{v}_1$ *is on the line being projected upon) and sends* $\boldsymbol{v}_2$ *to 0 (which makes sense given that* $\boldsymbol{v}_1 \perp \boldsymbol{v}_2$*).*

To summarize: expressing one and the same linear transformation in different bases (i.e. reference frames) is achieved by multiplying the corresponding matrix on the left and on the right by appropriate change of basis matrices. The key notion here is that of the change of basis matrix:

$$V = \left[\begin{array}{c|c|c} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_n \end{array}\right] \qquad \Leftrightarrow \qquad V\boldsymbol{e}_i = \boldsymbol{v}_i \quad \forall i \in \{1, \dots, n\}$$

for the change of basis matrix from the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$.

Fact 12. *All change of basis matrices are invertible.*

Indeed, if V changes from the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$, then its inverse should change back:

$$V^{-1}\boldsymbol{v}_i = \boldsymbol{e}_i \quad \forall i \in \{1, \dots, n\}$$

from the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ to $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$. So in particular the inverse should exist.

What if we wanted to change between two general bases, say from $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$? What is the change of basis matrix which achieves that? We can think about this as a two-step process:

$$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\} \xrightarrow{\text{change basis using matrix } V} \{\boldsymbol{e}_1, \dots, \boldsymbol{e}_n\} \xrightarrow{\text{change basis using matrix } W^{-1}} \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_n\}$$

So if we had a vector $\boldsymbol{a}$ expressed in the bases $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ and $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ and $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ as:

$$\boldsymbol{a} = x_1\boldsymbol{v}_1 + \cdots + x_n\boldsymbol{v}_n = y_1\boldsymbol{w}_1 + \cdots + y_n\boldsymbol{w}_n = z_1\boldsymbol{e}_1 + \cdots + z_n\boldsymbol{e}_n$$

then the Remark above says that the vectors of coefficients $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $\boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ are connected to the vector of coefficients $\boldsymbol{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$ by:

$$\boldsymbol{z} = V\boldsymbol{x} \qquad \text{and} \qquad \boldsymbol{z} = W\boldsymbol{y}$$

Setting the two formulas equal to each other implies that $\boldsymbol{x}$ and $\boldsymbol{y}$ are connected by:

$$\boxed{\boldsymbol{y} = W^{-1}V\boldsymbol{x}}$$

which is the way we change from the $\boldsymbol{v}$ basis to the $\boldsymbol{w}$ basis. In other words, the entries of the matrix $W^{-1}V$ give the coordinates of the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ when expressed in the basis $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$:

$$W^{-1}V = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{n1} & \cdots & c_{nn} \end{bmatrix} \quad \text{where} \quad \boldsymbol{v}_j = \sum_{i=1}^{n} c_{ij}\boldsymbol{w}_i \quad \forall j \in \{1, \ldots, n\}$$